WORKSHEET 6.8

Antiderivatives and the Basic Rules

Unit 6: Integration and Accumulation of Change

Key ideas

- $\int x^n\,dx = \dfrac{x^{n+1}}{n+1} + C$ for $n \neq -1$, and $\int \dfrac{1}{x}\,dx = \ln|x| + C$.
- Every derivative formula read backwards is an antiderivative formula: $\int \sec^2 x\,dx = \tan x + C$, and so on.
- An indefinite integral always includes $+C$.
- An initial condition fixes the constant. Two integrations need two conditions.

Worked examples

Example 1 · Term by term

Find $\int \left(6x^2 - 4x + 5\right) dx$.

Solution. Raise each power by one and divide by the new power.

$2x^3 - 2x^2 + 5x + C$.

Example 2 · Mixed functions

Find $\int \left(\dfrac{3}{x} - 2e^x + \sec^2 x\right) dx$.

Solution. Each term is a standard form.

$3\ln|x| - 2e^x + \tan x + C$.

Example 3 · With an initial condition

Given $f'(x) = 3\sqrt{x}$ and $f(4) = 10$, find $f(x)$.

Solution. $f(x) = 2x^{3/2} + C$, and $2(8) + C = 10$.

$f(x) = 2x^{3/2} - 6$.

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Practice

CORE

1. Find $\int x^5\,dx$.

2. Find $\int \left(4x^3 - 1\right) dx$.

3. Find $\int \cos x\,dx$.

4. Find $\int \frac{1}{x^2}\,dx$.

EXAM LEVEL

5. Find $\int (x+1)(x-2)\,dx$.

6. Find $\int \frac{x^2+4}{x}\,dx$.

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7. Find $\int \left(\csc^2 x + \dfrac{1}{1+x^2}\right) dx$.

8. Given $f''(x) = 6x$, $f'(1) = 5$ and $f(0) = 2$, find $f(x)$.

Question 9 · Free-response practice

A particle moves along a line with acceleration $a(t) = 6t - 4$. At $t = 0$ its velocity is 3 and its position is 1.

(a) Find the velocity $v(t)$.

(b) Find the position $x(t)$.

(c) Does the particle ever come to rest? Justify your answer.

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Worked solutions

1. $\dfrac{x^6}{6} + C$.
2. $x^4 - x + C$.
3. $\sin x + C$.
4. $-\dfrac{1}{x} + C$.
5. $\displaystyle\int \left(x^2 - x - 2\right) dx = \frac{x^3}{3} - \frac{x^2}{2} - 2x + C$.
6. $\displaystyle\int \left(x + \frac{4}{x}\right) dx = \frac{x^2}{2} + 4\ln|x| + C$.
7. $-\cot x + \arctan x + C$.
8. $f'(x) = 3x^2 + C_1$ with $3 + C_1 = 5$, so $f'(x) = 3x^2 + 2$. Then $f(x) = x^3 + 2x + 2$.
9. (a) $v(t) = 3t^2 - 4t + C$ with $C = 3$: $v(t) = 3t^2 - 4t + 3$.
   (b) $x(t) = t^3 - 2t^2 + 3t + 1$.
   (c) No. $3t^2 - 4t + 3 = 0$ has discriminant $16 - 36 < 0$, so $v(t)$ is never zero.

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